

DEPARTMENT OF MATHEMATICS ANNUAL REPORT 2006

PREAMBLE

The Department of Mathematics comprises over 50 established and temporary staff, based on both the City and Tamaki campuses. The number of equivalent full-time students totalled about 710 in 2006. This number is expected to increase in 2007. The Department remains one of the largest in the University of Auckland, and offers courses at all levels for students in several Faculties.

Highlights for the Department in 2006 were:

- Prof. Ivan Reilly was made an Officer of the NZ Order of Merit in the New Year Honours.
- Prof. James Sneyd won the James Cook Fellowship.
- Prof. Marston Conder was appointed to the Chair of the Academy of the Royal Society of New Zealand
- A/Prof. Paul Bonnington's successful multi-million dollar bid to TEC to build digital science capability in New Zealand.
- A/Prof. Rod Gover's and Dr Paul – Andi Nagy's success in winning a Marsden Fund grant this year for a project on "Natural operators, special structure and conformal techniques". Also Dr Mike Meylan's Marsden grant with Vernon Squire has been extended for another two years, and will include a post-doctoral fellow.
- A/Prof. Bill Barton was elected as Vice-President of the International Commission on Mathematical Instruction.
- Dr. Warren Moors gave a plenary address at the 10th (5-yearly) Topology Symposium in Prague.
- Prof. James Sneyd won funding for a University of Auckland Post-doctoral fellowship. This is the third year in a row that someone from the department has won such a fellow.
- The Department now hosts four Top Achiever Doctoral Students: Eyal Loz and Inga Wang joining Elan Gin and Erin Higgins.
- The Department hosted University Foundation visitors Profs. Hyman Bass and Deborah Ball from Michigan University.

Associate Professor Eamonn O'Brien was promoted Professor of Mathematics. Dr Philip Sharp was promoted Senior Lecturer over the bar, Dr Sina Greenwood was promoted to Senior Lecturer, and Dr Allison Heard was promoted over the Senior Tutor bar in the 2006 round.

The Department hosted several post-doctoral students at various stages of the year: Drs Richard Evans (NZIMA), Edward Huang (NZ Science & Technology), Hyuck Chung (NZ Science & Technology), Caroline Poisard (HOD), Laura Ciobanu (University of Auckland), and Paul-Andi Nagy (University of Auckland).

After 40 years service we farewelled Mr Roy Swenson who has retired. Roy has been a Senior Tutor contributing to the first year programme and managing the tutors and markers. We will miss his behind-the-scenes work and institutional knowledge. Also leaving are Mr David Thomson (another long-serving contributor to the first year programme), Mr Viliami Latu (a tutor in the Maori-Pasifika section), and Dr. Jiling Cao (to a permanent appointment at AUT). The recent announcement by Associate Professor Jozef Siran that he will be leaving to take up a professorship at The Open University in the UK was greeted with dismay that he is leaving on the one hand and congratulations for his promotion on the other. In the general staff area we farewelled Ms Patricia Rood and Ms Kristin Osborne, and warmly welcomed Ms Adina Nagy, Ms Rowena-Knight-Brown, and Ms Julia Novak (who is also taking on some teaching duties).

The Department was the subject of an external review during 2006. This major undertaking has provided ideas for development as well as affirming the standing and functioning of the department.

I. DEVELOPMENTS IN TEACHING AND LEARNING

The Department provides teaching in courses for students in several Faculties, especially Arts, Business & Economics, Engineering, and Science. A major part of this is service teaching as mathematics provides a crucial part of many different majors.

For the 2006 academic year student enrolments totalled approximately 707 EFTS, including 650 undergraduate (630 on City and 20 on Tamaki Campuses), and 57 graduate EFTS, including 17 doctoral EFTS. International students make up 22% of our enrolments. This represented a significant drop in EFTS, that had two main causes. First was the full-time student load restructuring from 7 to 8 courses per year. Since a large proportion of Mathematics enrolments are as a service subject, Mathematics did not pick up its share of extra courses, students

choosing to take their extra courses in their major subjects. In addition, a restructuring in the Faculty of Business & Economics saw a mathematics course dropped from their programme with a subsequent loss of about 70 EFTS.

After the 2005 revision of all undergraduate courses including delivery and assessment procedures, 2006 saw the first run through of the new programme. The adoption of a single text-book package for 100- and 200-courses met with some resistance as students needed to learn how to reference mathematics outside lecture notes, but the adoption of MatLab as a computer platform for all undergraduate mathematics students went smoothly and was well accepted. This year also saw the trialing of Tablet technology to record lectures in real-time and make the recordings available through Cecil. This proved to be very successful and is being further developed in 2007.

The new General Education course MATHS 190, along with the beginning course MATHS 101 were both chosen by sufficient students to ensure their continuance. The increasingly successful year-long version of MATHS 150 for top secondary students also continues to grow and attract the very best students into the University of Auckland.

Although the planning for proper space to receive and tutor mathematics students in the 303 Building Renovations is still in place, we are anxious that it be ready soon as our need for proper tutorial facilities continues to grow.

This year has seen our attention focused on ensuring that there is a coherent pathway for majors in Mathematics or Applied Mathematics. Several courses are being revised to be more accessible for all majors.

The mathematics courses for the PGDipEd(Sec) programme were taught in collaboration with the Department for the last time, Faculty of Education reorganization involving the canceling of these unique and successful courses.

The programme of Public Lectures started in 2005 continued extremely successfully, and now represent a major outreach activity that brings the public to the university. The hosting of the Great NZ Origami Show that undertook a successful tour of the country also raised the department's and the university's profile.

II. OTHER STUDENT MATTERS

Department of Mathematics pass rates have improved significantly, particularly in our large 100- and 200-level classes. Our efforts are now directed at the more advanced undergraduate classes to bring them into line with university norms.

We are aware of the thrust of the University's Strategic Plan to dramatically increase graduate numbers. Our PhD enrolments continue to increase steadily, and the new initiatives by International Office in Pakistan are welcomed. We are delighted to have already enrolled several students and are engaged in negotiations with others.

In 2006 we had 5 doctoral completions and 10 new doctoral enrolments, bringing our current number of doctoral students to 25.

Scholarships and Prizes

Prizes are:

Senior Scholarship in Mathematics	Luke Ford
Annual Prize in Mathematics	Daniel Harris
Senior Scholarship in Applied Mathematics	Jamie Kidd
Senior Prize in Mathematics	Anmei Shen
Senior Prize in Mathematics	Helen Broome
Senior Prize in Mathematics	Mark Hashimoto
Senior Prize in Mathematics	Edward Kelly
Senior Prize in Mathematics	Heather Macbeth
Senior Prize in Mathematics	Claire Radley
Senior Prize in Mathematics	Natalie Rolfe
Senior Prize in Mathematics	Yany Yu
Senior Prize in Mathematics	Won Min Song
The Margaret Morton Memorial Prize	Dinushi Jayasuriya
Collins Prize	Simon Marshal
Mathematics Education Prize	Stephanie Ng
Mathematics Education Technology Prize	Johanna McHardy

Summer Scholarships

Nineteen students were engaged in research projects over the summer 2006-7, with seven gaining scholarship support from the Department, and twelve from the Faculty of Science. The basic aim of the scheme is to introduce senior undergraduates to mathematical research. The programme of summer scholarships continues to be invaluable to both students and staff.

Aldis Scholarships

These scholarships are discussed in more detail in Section IV. They are awarded to very able students from Decile 1 and 2 High Schools to enable them to enroll in a Summer Season course in Mathematics.

Student/Staff Liaison Committee

Mr Greg Oates continued to convene this Committee. The meetings continue to attract a full room of student and staff representatives with a range of important issues discussed.

Tutor Training

The Department of Mathematics has long organized a tutor training course under the direction of Mr Greg Oates. This course is a model for the CAD of a department-oriented course, and its certificate has been adopted by the Faculty of Science.

III. EQUAL OPPORTUNITIES

While mathematical ability is represented equally amongst all sectors of the population, women are still in the minority amongst our students with 38% being female. Furthermore, Maori and Pasifika students remain underrepresented with 2.5% and 7.5% respectively. This situation carries on to the Academic staff of the Department. For example, only one fifth of permanent Academic staff are female. Typically, few of the applicants for new academic positions are female.

Over half of students enrolled in Mathematics are of ethnic Asian origin, so the two permanent staff members who are themselves of Chinese origin are used within the Department as liaison and support for these students. We also have many tutors who are able to, and frequently do, conduct their tutoring sessions with individuals or in groups in the Mandarin language where this is appropriate.

A similar situation applies to Maori and Pasifika students although the programme there is more formalised. Through the Tuakana programme and the efforts of three staff members (a Maori, a Tongan and a Samoan), students from these origins are able to study together in a culturally supportive environment and receive support in their language in many cases. A scholarship programme to bring Maori and Pasifika students into Year 1 university study (the Aldis programme) continues. Sixteen Aldis scholars were in the programme in 2006, and the pass rates for these scholars continue to exceed the norms for either the Faculty or the university as a whole.

The Department makes special efforts both with the mathematically able and those needing extra support. The MAX programme for talented students still in their final year of school is growing considerably, and the Department continues to have involvement in the International Mathematics Olympiad programme. For students needing support or a second chance, the Mathematics Department continues its strong support of the Tertiary Foundation Certificate.

The EEdO activities of the Department now include a comprehensive suite of research projects: the MEP programme in Manukau low-decile schools, a programme in Northland schools tracking good mathematics students from Intermediate age, and research into the learning of mathematics by students who do not have English as a first language.

IV. TREATY OF WAITANGI

As noted in last year's report, the Department of Mathematics has a long history of serious attention to responsibilities under the treaty. Its on-going Maori and Pasifika programme led by Dr Sina Greenwood assisted by Mr Garry Nathan and Mr Viliami Latu, has been successful in attracting and retaining Maori and Pasifika students - although the numbers are still acknowledged to be low. The programme continues to build a group of trained mentors and role-models, and uses them in attracting new students and supporting existing ones.

A significant part of this effort is the awarding of Aldis Scholarships. Since this programme is also a service function in that Aldis scholars enter programmes in many different Faculties and Departments other than Mathematics, we are delighted that this is now funded centrally. We retain a commitment to administering the programme.

V. RESEARCH

The Department continues to perform as the top Mathematics Department in New Zealand, and on a par with the best in Australia. Yet again the Department does well in the Marsden, ISAT, and TLRI Funding rounds; yet again a member of the Department won a James Cook Fellowship; yet again the Department's publication list includes the top journals in the field and a wide variety of sub-areas within mathematics are included; yet again members of the Department are invited plenary speakers at international conferences; yet again this Department hosts NZIMA programmes; and yet again our graduate students win significant awards and scholarships.

It is a measure of the strength of the Department of Mathematics' research profile that we continue to host international conferences and seminars, with more than two seminars per week on average. Of further note was the signing of two single author book contracts with major international scientific publishers.

The influx of some PBRF-generated income directly to the Department has been widely welcomed this year so that our visitor and visiting programme can be maintained. Internally administered grants have been awarded on a research-output basis.

It has become traditional in our Annual Report to highlight one area as an example of our research activity. In 2006 we highlight the work of A/Prof. Rod Gover and Dr. Paul-Andi Nagy. Their work on special geometric structures and conformal geometry has importance in the mathematics underlying aspects of theoretical physics. More than a dozen papers have resulted from the work in 2006, and A/Prof. Gover was invited to the top research institute in France.

The publications, conference activity, international visitor list, and service to the field contributions noted below speak for themselves.

PUBLICATIONS, CREATIVE WORKS AND OTHER RESEARCH OUTPUTS

(A) Book - Authored

GAULD D., *Differential Topology: an introduction*. Mineola, New York, Dover, 241 pages, 2006. (Corrected reprint of original published in 1982 by Marcel Dekker)

(B) Book – Chapter

ALEKSEEV, V., BARTON, B., & KNIJNIK, G. 'Other Conventions in Mathematics & Mathematics Education'. In F. Leung, K. Graf, & F. Lopez-Real, *Mathematics Education in Different Cultural Traditions: A Comparative Study of East Asia and the West*. The 13th ICMI Study. New York, Springer Science and Business Media, p.567-580

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(C) Journal articles – refereed articles in scholarly journals

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(E) Conference publications – full written papers, refereed

CHRISTIAN, R.D., FELLOWS, M., ROSAMOND, F., SLINKO, A.M. 'On Complexity of Lobbying in Multiple Referenda', Ulle Endriss and Jerome Lang(ed.), *Proceedings of the 1st International Workshop on Computational Social Choice, Amsterdam, 6 - 8 December, 2006*, p.87-96, <http://staff.science.uva.nl/~ulle/COMSOC-2006/comsoc2006.pdf>

CHUNG, H., EMMS, G., MCGUINNIGLE, K., DODD, G. 'Improving the impact insulation of light timber floors', *Noise of Progress, Proceedings of 1st Joint Australasian Acoustical Conference, Christchurch, New Zealand, November 20-22, 2006*, 2006, p.-

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FOX, C., CHUNG, H., DODD, G., SCHMID, G., EMMS, G., MCGUINNIGLE, K. 'Calibrated modelling of low-frequency vibration in light weight timber floors', *Inter Noise 2006, I-N 2006 Proceedings, Honolulu, Hawaii, USA, December 3-6, 2006*, 2006, p.-

HAVAS, G., LEEDHAM-GREEN, C. R., O'BRIEN, E.A., SLATTERY, M.C. 'Computing with elation groups', Hulpke et al.(ed.), *Finite Geometries, Groups and Computation, Colorado, September 2004*, 2006, p.95-102

HONG, Y.Y., THOMAS, M.O.J. 'Factors influencing teacher integration of graphic calculators in teaching', *Proceedings of the 11th Asian Technology Conference in Mathematics, Hong Kong, December, 2006*, p.234-243

MCCABE-DANSTED, J.C., PRITCHARD, G., SLINKO, A.M. 'Approximability of Dodgson's rule', in Ulle Endriss and Jerome Lang(ed.), *Proceedings of the 1st International Workshop on Computational Social Choice (COMSOC--2006), Amsterdam, 6-8 December, 2006*, p.331-344, <http://staff.science.uva.nl/~ulle/COMSOC-2006/comsoc2006.pdf>

MERRIMAN, L.V. 'Using media reports to develop statistical literacy in year 10 students', *7th International Conference on Teaching Statistics*, A. Rossman & B. Chance(ed.), *Working Cooperatively in Statistics Education, Salvador, Bahia, Brazil, 2-7 July 2006*, 2006, p.-, <http://www.stat.auckland.ac.nz/~iase/publications.php>

MILLER-REILLY, B.J. 'Adults Returning to Mathematics: Affective responses to three different teaching approaches', *The 13th International Adults Learning Mathematics Conference (ALM13)*, Valerie Seabright(ed.), *Crossing Borders: Research, Reflection and Practice in Adults Learning Mathematics*, Queen's University Belfast, Northern Ireland, 16 - 20th July 2006, 2006, p.-

O'BRIEN, E.A. 'Towards effective algorithms for linear groups', *Finite Geometries, Groups and Computation*, Hulpke et al.(ed.), *Finite Geometries, Groups and Computation, Colorado, September 2004*, 2006, p.163-190

PAVLOV B., 'A Dirichlet-to-Neumann approach to resonance gaps and bands of periodic networks Contemporary mathematics', 412 (2006) *Proceedings of the Conference: Operator Theory and Mathematical Physics*, Birmingham, Alabama, April 2005, pp. 151-169.

PFANNKUCH, M. 'Informal inferential reasoning', *7th International Conference on Teaching Statistics*, A. Rossman & B. Chance(ed.), *Working Cooperatively in Statistics Education*, Salvador, Bahia, Brazil, 2-7 July 2006, 2006, p.-, <http://www.stat.auckland.ac.nz/~iase/publications.php>

POISARD, C., 'The notion of carried-number, between the history of calculating instruments and arithmetic'. In P. Grootenboer, R. Zevenbergen, & M. Chinnappan (Eds.), *Identities, cultures and learning spaces* (Proceedings of the 29th annual conference of the Mathematics Education Research Group of Australasia, Canberra, 1-5 July, Vol 2, pp. 416-423). Adelaide, 2006

ROBINSON, M., BARTHOLOMEW, H. 'There's more to maths than doing sums', *Proceedings of the Third International Conference on the Teaching of Mathematics*, Istanbul, Turkey, 30th June - 5th July 2006, 2006, p.-

SIRAN, J. 'Regular maps on a given surface - a survey', *6th Czech and Slovak International Symposium on Graph Theory*, R. Thomas et al.(ed.), *Topics in Discrete Mathematics*, Prague, July 10-15, 2006, 2006, p.591-609

STEWART, S., THOMAS, M.O.J. 'Process-Object Difficulties In Linear Algebra: Eigenvalues and Eigenvectors', *Proceedings of the 30th Conference of the International Group for the Psychology of Mathematics Educ*, 5, Prague, Czech Republic, July, 2006, p.185-192

STEWART, S., THOMAS, M.O.J. 'Student thinking about eigenvalues and eigenvectors: Formal, symbolic and embodied notions', P. Grootenboer, R. Zevenbergen & M. Chinnappan(ed.), *Proceedings of the 29th Annual Conference of the Mathematics Education Research Group of Australasia*, 2, Canberra, Australia, July, 2006, p.487-495

TER ELST, A.F.M., ROBINSON, D.W., SIKORA, A., ZHU, Y.P. 'Dirichlet forms and degenerate elliptic operators', *Partial differential equations and functional analysis, Oper. Theory Adv. Appl.*, 168, Delft, The Netherlands, 29-11-2004 -- 1-12-2004, 2006, p.73-95

THOMAS, M.O.J. 'Developing versatility in mathematical thinking', *A Festschrift for Eddie Gray and David Tall*, A. Simpson(ed.), *Retirement as Process and Concept*, Charles University, Prague, Czech Republic, July, 2006, p.223-241

THOMAS, M.O.J. 'Teachers Using Computers In Mathematics: A Longitudinal Study', C. Hoyles, J.-B. Lagrange, L. H. Son, & N. Sinclair(ed.), *Proceedings of the Seventeenth ICMI Study Conference "Technology Revisited"*, Hanoi, Vietnam, December, 2006, p.545-554, CD version of Proceedings

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Conference Contribution – Abstract

MILLER-REILLY, B.J. 'Pre-Calculus Mathematics: Students' affective responses to different teaching approaches', *The 3rd International Conference On The Teaching Of Mathematics (at the undergraduate level) (NCTM3)*, Istanbul, Turkey, 30th June - 5 July 2006, 2006, p.-

For other conference contributions – see Section VI. Staff leave and Conferences

(F) Scholarly Reviews and Comments (published in journals of repute)

DIKSHIT, G.D.: Bilgin, Tunay, Matrix transformation on the sequence space $c_0(p,s)$. Review in *Zentralblatt* 1025.40005

DIKSHIT, G.D.: Gupta, Vijay, Degree of approximation to function of bounded variation by Bêzier variant of MKZ operators. Review in *Zentralblatt* 1037.41013

DIKSHIT, G.D.: Abel, Ulrich, On the Lagrange remainder of the Taylor formula. Review in *Zentralblatt* 1056.41026

DIKSHIT, G.D.: Pinkus, Alan Negative theorems in approximation theory. Review in *Zentralblatt* 1049.41013

- DIKSHIT, G.D.: Portilla, Ana; Quintana, Yamilet; Rodriguez, Jos[^]e M.; Touris, Eva :Weierstrass' theorem with weights. Review in *Zentralblatt* 1045.41005
- DIKSHIT, G.D.: Song, Wenhua; Liu, Guofen The $|\mathfrak{f}|$ -width in locally convex spaces. Review in *Zentralblatt* 1046.41009
- DIKSHIT, G.D.: Navascuês, M.Antonia; Sebastian, M.Victoria , Some results of convergence of cubic spline fractal interpolation functions. Review in *Zentralblatt* 1053.41013
- DIKSHIT, G.D.: Walczak, Z On the rate of convergence for modified Baskakov operators. Review in *Zentralblatt* 1058.41007
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- DIKSHIT, G.D.: Nikolov, Geno An extension of an inequality of I. Schur. Review in *Zentralblatt* 1075.41010
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(G) Technical Reports

CHRISTIAN, R.D., CONDER, M.D.E., SLINKO, A.M. 'Flippable Pairs and Subset Comparisons in Comparative Probability Orderings and Related Simple Games', *Cahier*, 15, The Centre for Interuniversity Research in Qualitative Economics (CIREQ), Montreal, Canada, 2006, <http://www.cireq.umontreal.ca/Profpublications/15-2006-cah.pdf>, 1-24

CHRISTIAN, R.D., FELLOWS, M., ROSAMOND, F., SLINKO, A.M. 'On Complexity of Lobbying in Multiple Referenda', *Cahier*, 13, Centre for Interuniversity Research in Qualitative Economics (CIREQ), Montreal, Canada, 2006, <http://www.cireq.umontreal.ca/Profpublications/13-2006-cah.pdf>, 1-7

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GARTSIDE, P. AND GREENWOOD, S. 'Brunnian links'. *Fundamenta*, (in press)

KORAY, S., SLINKO, A.M. 'Self-Selective Social Choice Functions', *Cahier*, 18, The Centre for Interuniversity Research in Qualitative Economics (CIREQ), Montreal, Canada, 2006, <http://www.cireq.umontreal.ca/Profpublications/18-2006-cah.pdf>, 1-22

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Mathematics Department Research Report Series

- 555: Four dimensional conformal C-spaces, by Rod Gover and Paul-Andi Nagy
- 554: Arnold's analytic perturbation for the polar equation, by Samuel Dillon
- 553: Some results on Linearly Lindelöf spaces, by Hongfeng Guo, Houli Jiang
- 552: On algebraic torsion forms and their spin holonomy algebras, by Niels Bernhardt and Paul-Andi Nagy
- 551: Approximability of Dodgson's rule, by John C McCabe-Dansted, Geoffrey Pritchard, Arkadii Slinko
- 550: Generalized Convexity and Inequalities, by G D Anderson, M K Vamanamurthy, and M Vuorinen
- 549: Holomorphic forms of geometrically formal Kähler manifolds, by Jean-Francis Grosjean and Paul-Andi Nagy, to appear in *Proceedings of the London Mathematical Society*
- 548: Boundary condition at the junction, by M Harmer, B Pavlov, A Yafyasov
- 547: On the manipulability of proportional representation, by Arkadii Slinko and Shaun White
- 546: On the Order Hereditary Closure Preserving Sum Theorem, by Jianhua Gong and Ivan L Reilly
- 545: Character Tables of Parabolic Subgroups of the Chevalley Groups of Type G_2 , by Jianbei An and Shih-chang Huang
- 544: Dade's invariant conjecture for the Chevalley groups of type G_2 in the defining characteristic, by Shih-chang Huang

OTHER MATTERS RELATED TO RESEARCH

Departmental Seminars by Visitors, Honorary Research Fellows, Staff and Research Students

- | | | |
|------------|--|--|
| 15-12-2006 | Dr. Steven Galbraith
(Royal Holloway, University of London) | Some computational problems arising in public key cryptography |
| 15-12-2006 | Dr. Paul-Andi Nagy
(Department of Mathematics) | Supersymmetry and PSU(3) structures |

14-12-2006	Prof. Boris Altshuler (Columbia University, NY)	From Quantum Chaos to Anderson Localization
08-12-2006	Dr. Jonathan Kress (University of New South Wales)	Superintegrable systems and algebraic varieties
7-12-2006	Prof. Richard L. Hall (Concordia University, Montreal, Canada)	Geometric spectral inversion
1-12-2006	Dr. Howard Cohl (Department of Mathematics)	Green's functions for the 3D Laplace equation
30-11-2006	Prof. Richard Hall (Concordia University, Montreal, Canada)	Systems of Identical Particles
28-11-2006	Prof. John Grue (University of Oslo, Norway)	Fully Non-linear Simulations of Rogue Waves in Three Dimensions and Comparison with Experiments using Particle Image Velocimetry
28-11-2006	Stephen Budden (Department of Mathematics)	Quandles and Knots
24.11.2006	Callum Sleigh (Department of Mathematics)	Harmonic analysis on locally compact abelian groups
15-11-2006	Dr. Martin Wechselberger (University of Sydney)	Giant Squid - Hidden canard -- the Geometry of the Hodgkin-Huxley
14-11-2006	Dr. Sina Greenwood (Department of Mathematics)	Characterising continuous functions
13-11-2006	Prof. Abdolaziz Abdollahi (Shiraz University, Iran)	The numerical range of a composition operator
10-11-2006	Dr. Paul-Andi Nagy (Department of Mathematics)	Kähler manifolds
8-11-2006	Prof. Pavel Exner (Doppler Institute, Prague)	Generalized graphs - or what happens if a quantum particle has to change its dimension
8-11-2006	Sione Ma'u (Indiana University)	Pluripotential theory for convex bodies in $\mathbb{R}^n$
7-11-2006	Dr. Jiling Cao (Auckland University of Technology)	Baire spaces, Tychonoff powers and the Vietoris topology
6-11-2006	Dr. Colva Roney-Dougal (University of St Andrews)	Classification and Construction of Finite Permutation Groups
3-11-2006	Prof. Pavel Exner (Doppler Institute, Prague)	Leaky graphs - what they are, and their spectral and resonance properties
31-10-2006	A/Prof. Rod Gover (Department of Mathematics)	Symmetry operators for the Laplacian on E^n (IV)
25-10-2006	Prof. Pavel Exner (Doppler Institute, Prague)	Ideal graphs - their nontrivial aspect, or what is the meaning of the vertex coupling?
25-10-2006	Prof. Derek W. Robinson (Australian National University)	Commutator theory on Hilbert space
17-10-2006	Dr. Sina Greenwood (Department of Mathematics)	750 Assignment 3, question 4, 2006, with typo
16-10-2006	Julia Novak (Royal Holloway and UoA)	Group key distribution patterns
13-10-2006	A/Prof. Rod Gover (Department of Mathematics)	Symmetry operators for the Laplacian on E^n (III)
11-10-2006	Dr. Ilze Ziedins (University of Auckland)	When networks go bad...

9-10-2006	Prof. Peter Brooksbank (Bucknell University)	Algorithms for finitely-dimensional modules
22-10-2006	Mark Nelson and Harvi Sidhu (Uni University of Wallongong and Australian Defence Force Academy)	Compost Piles and wastewater Treatment
06-10-2006	A/Prof. Rod Gover (Department of Mathematics)	Symmetry operators for the Laplacian on E^n (II)
05-10-2006	Sepideh Stewart (Department of Mathematics).	Communicating linear algebra using the embodied, symbolic and formal worlds of mathematics
01-10-2006	Dr. Philip Sharp (Department of Mathematics)	Using symplectic methods for long simulations of the Solar System
29-09-2006	A/Prof. Rod Gover (Department of Mathematics)	Symmetry operators for the Laplacian on E^n (I)
22-09-2006	Niels Bernhardt (Department of Mathematics)	Spin structures on Riemannian manifolds (II)
21-09-2006	Dr. Caroline Poisard (Department of Mathematics)	A comparison between New Zealand and French education.
19-09-2006	Dr. Ernie Kalnins (University of Waikato)	Separation of Variables
18-08-2006	Dr. Howard Cohl (Department of Mathematics)	Separation of variables from linear partial differential equations from a Lie group and algebraic perspective (II)
18-09-2006	Dr. Frank Himstedt (Munich University of Technology)	Alternating squares of modules for cyclic groups
15-09-2006	Greg Oates (Department of Mathematics)	'The Great Origami Maths and Science Show: What's up in Origami and Mathematics Down- under?', Maths Education Seminar Series,
15-09-2006	Niels Bernhardt (Department of Mathematics)	Spin structures on Riemannian manifolds (I)
12-09-2006	Niels Bernhardt (Department of Mathematics)	Clifford algebras and spinors (II)
01-09-2006	Niels Bernhardt (Department of Mathematics)	Clifford algebras and spinors (I)
31-08-2006	Prof. Alan McIntosh (ANU, Canberra)	The Square Root Problem of Kato for Elliptic Operators: Survey, Solution and Sequel
29-08-2006	Dr. Mark McGuinness (School of Maths, Stats, and Computer Science, Victoria University of Wellington)	Modelling Platelets in Antarctic Sea Ice
25-08-2006	Dr. Paul-Andy Nagy (Department of Mathematics)	Principles of separation of variables in the spectrum of the Laplacian
21-08-2006	Prof. Helcio R. B. Orlande (Federal University of Rio de Janeiro, Brazil)	Identification of heat flux imposed by an oxyacetylene torch
15-08-2006	Niels Bernhardt (Department of Mathematics)	On algebraic torsion forms and their spin holonomy algebras
09-08-2006	Hyman Bass and Deborah Ball (University of Michigan)	Improving teaching and teaching proving
07-08-2006	Prof. Hyman Bass (University of Michigan)	Rationality of the Zeta Function of a Finite Graph
11-08-2006	Dr. Howard Cohl (Department of Mathematics)	Separation of variables from linear partial differential equations from a Lie group and algebraic perspective (I)
28-07-2006	Prof. Shouli Jiang (Shandong University)	The cardinality of a first countable compact space

27-07-2006	Dr. Bjorn Sandstede (University of Surrey)	Dynamics of Spiral Waves
26-07-2006	Dr. Bjorn Sandstede (University of Surrey)	A gentle introduction to nonlinear PDEs
25-07-2006	Prof. William Newman (Departments of Mathematics, Physics and Astronomy, Earth & Space Sciences, UCLA)	Earthquakes, Forest Fires, Gang Warfare, and Related Topics
29-06-2006	Dr. Dmitry Malinin (USP Fiji)	Finite Galois stable subgroups of GL_n
20-06-2006	Dr. Emily Stone (The University of Montana- Missoula)	Detecting Spillover: A dynamical systems modeling approach to glutamatergic synaptic signaling
12-06-2006	Dr. Mathieu Baillif (Université de Genève)	Homotopy in non-metrizable manifolds
12-06-2006	Prof. Jon Carlson (University of Georgia)	The poset of elementary abelian p -subgroups of rank at least two
07-06-2006	Dr. Abdul Mohamad (Sultan Qaboos University)	Probabilistic Quasi-Metric Spaces
07-06-2006	Tara Bonda (Department of Mathematics)	Alternating and symmetric quotients of certain triangle groups
06-06-2006	Prof. Peter Nyikos (University of South Carolina)	Hereditarily normal manifolds
31-06-2006	Prof. Peter Nyikos (University of South Carolina)	Subsequentiality and sequential compactness
29-05-2006	Prof. Peter Nyikos (University of South Carolina)	Kurepa families and locally compact, countably compact spaces
29-05-2006	Dr. Peter A. Streufert (University of Western Ontario)	Products of Relative Probabilities
24-05-2006	Prof. Peter Nyikos (University of South Carolina)	Stone-Ćech remainders of discrete spaces: some old problem
22-05-2006	Prof. Peter Nyikos (University of South Carolina)	Sequential compactness (and lack thereof) in countably compact spaces
22-05-2006	Dr. Robert Jajcay (Indiana State University)	Classifying regular Cayley maps
22-05-2006	Dr. Arkadii Slinko (Department of Mathematics)	Abstract Convex Geometries and Decision Rules
22-05-2006	Prof. Peter Nyikos (University of South Carolina)	In and around the M_1 - M_3 problem
18-05-2006	Dr. Rick Beatson (University of Canterbury)	Radial basis function theory and applications
17-05-2006	Dr. Richard Evans (Department of Mathematics)	Characterising strong convergence
10-05-2006	Dr. Simon Chiossi (Humboldt University, Berlin)	Solvable extensions of nilpotent Lie groups with G_2 structures
10-05-2006	Dr. Tsukasa Yashiro (Department of Mathematics)	Surface diagrams with two triple points
10-05-2006	Simon Marshall (Department of Mathematics)	Bound States of Schrodinger Operators
08-05-2006	Dr. Walter Bossert (University of Montreal)	Arrow's theorem
03-05-2006	Dr. Warren Moors (Department of Mathematics)	When is a compact convex metrizable?
02-05-2006	Dr. Gareth Vaughan	The scattering of ice-coupled waves by an arbitrary

	(Otago University)	ice sheet
02-05-2006	Mr. Ronald Begg (University of Canterbury)	Some problems related to cell-growth modeling
13-04-2006	Prof. Jochen Bruening (Humboldt-Universität zu Berlin)	Nonparabolic Dirac Systems
26-04-2006	Dr. Ben Martin (University of Canterbury)	Lattices, local fields and automorphism groups of trees
26-04-2006	Dr. Colin Guillarmou (University of Nice and the ANU)	Determinants, zeta functions, and Weyl asymptotics in hyperbolic geometry.
12-04-2006	Dr. Sean Cleary (City University of New York)	Thompson's group F and its remarkable Cayley graph
12-04-2006	Prof. David Gauld (Department of Mathematics)	Exploring the group of homeomorphisms of non-metrisable manifolds
06-04-2006	Prof. Derek Robinson (Australian National University)	Evolution equations and elliptic operators
05-04-2006	Dr. Tsukasa Yashiro (Department of Mathematics)	Cell-complexes for surface diagrams
03-04-2006	Dr. Murali Agastya (University of Sydney)	The core of an n-person game
03-04-2006	Dr. Richard Evans (Department of Mathematics)	Self-bumping and how to avoid it
31-03-2006	Sepideh Stewart (Department of Mathematics)	University students' understanding of linear algebra concepts, Graduate Research Interactive Discussion,
29-03-2006	Dr. Roger Fenn (University of Sussex)	Long (but not too long) Knots
27-03-2006	Matthew Auger (Department of Mathematics)	Coxeter Groups and Hyperbolic 4-Manifolds
14-03-2006	Dr. Helmut Podhaisky (Martin-Luthe-University Halle-Wittenberg)	Programming, puzzles and games
13-03-2006	Dr. Konstantin Pankrashkin (Humboldt-University of Berlin)	Spectral duality between quantum and combinatorial graphs
13-03-2006	Prof. R. Bruce Richter (University of Waterloo)	The Jordan Curve Theorem Revisited
09-03-2006	Dr Hinke Osinga (University of Bristol)	Global Manifolds: Computations & Crochet
08-03-2006	J. E. Paterson (Department of Mathematics)	A search for underlying mechanisms in a model of teacher change', Graduate Research Interactive Discussion
07-03-2006	Prof. Bernd Krauskopf (University of Bristol)	Compound laser modes of mutually delay-coupled lasers
02-03-2006	Prof. Michael Eastwood (University of Adelaide)	Higher symmetries of the Laplacian
23-02-2006	Prof. Ruth Kellerhals (University of Fribourg)	Some aspects of higher dimensional geometry
23-02-2006	Dr. Malte Peter (University of Bremen, Germany)	Modelling and homogenisation of chemical degradation mechanisms in porous media with evolving microstructure
20-02-2006	Prof. Ruth Kellerhals (University of Fribourg)	Polylogarithms and non-Euclidean volume

13-02-2006	Dr. Thierry Hild (University Fribourg)	Small volume cusped hyperbolic orbifolds
01-02-2006	Dr. Ernie Kalnins (University of Waikato)	Superintegrability and polynomial algebras (An Introduction)
23-01-2006	Prof. John Conway (Princeton University)	Integral lexicographic codes
19-01-2006	A/Prof. Mike Thomas (Department of Mathematics)	Versatile Mathematical Thinking: Perspectives on Symbols and Concepts

OTHER MATTERS RELATED TO RESEARCH ACTIVITIES

(a) PhD Students

The following students completed their PhD degrees during the year.

Name	Thesis Topic/Title	Supervisor(s)
Renu Choudhary	Convergence of Lyapunov functions along trajectories of nonexpansive semigroups: generic convergence and stability	A/Prof. B. Calvert
Alan De Los Santos	An investigation of students' understanding and representation of derivative in a graphic calculator – mediated teaching and learning environment	A/Prof. M. Thomas
Gareth Hegarty	Stabilisation of non-linear beams	Dr. S. Taylor
Joséf Šilhan	Invariant differential operators in conformal geometry	A/Prof R Gover,
David Welch	Stochastic models of population dynamics	Dr. G Nicholls, Prof. A Rodrigo (Biology), Dr. W Solomon

Current PhD students

Name	Thesis Topic/Title	Supervisor(s)
Willy Alanguai	Mathematics and Culture	A/Prof. B. Barton
Matthew Auger	Methods for proving certain groups are infinite	Prof. M. Conder, Prof. E. O'Brien
Niels Bernhardt	Parallel spinors, torsion forms and some classes of geometric structures.	A/Prof. R Gover, Dr. P. Nagy
Rachel Bevan	Estimating and modeling rates of evolution	Dr. D. Bryant, Dr. F. Lang
Trevor Bruen	Statistical Phylogenetics	Dr. D. Bryant
Stephen Budden	Quandles and knots	Prof. D. Gauld, Dr. S. Greenwood, A/Prof. J. Sirán
Damien Burns	Linear Groups	Prof. E. O'Brien A/Prof. J. An

Howard Cohl	Symmetries of Inverse Partial Differential Operators	A/Prof. R. Gover, Dr. T. ter Elst
Greg Ewing	Migration-genealogy coalescent processes	Dr. G. Nicholls, Prof. A. Rodrigo (SBS)
Elan Gin	A mathematical model of the mechanisms underlying saliva secretion	Prof. J. Sneyd
David Godfrey	Learning Algebra: The Arithmetic-Algebra Interface	A/Prof. M. Thomas, Dr K Irwin (Education)
Maria Goodier	Implementation of Novel Numerical Methods	Prof. J. Butcher, Dr. S. Wang
Yousaf Habib	Structure preserving algorithms for differential equations	Prof. J. Butcher, Dr. R. K. Chan
Erin Higgins	Modelling calcium transients in cardiac myocytes	Prof. J. Sneyd
Barbara Kensington-Miller	Professional Development of mathematics teachers in low decile 2 schools	A/Prof B Barton, Prof. S. McNaughton (Education)
Alison Kohout	Wave scattering in Antarctica	Dr. M. Meylan
Thomas Lepage	Evolutionary rate models	Dr. D. Bryant, P. Tupper
Paul Leys	Modelling schools of fish	Dr. P. Sharp, Prof. J. Sneyd
Eyal Loz	Degree Diameter Problem	A/Prof. P Bonnington A/Prof J Sirán
Shawn Means	Calcium Dynamics in Excitable Biological Cells	Prof. J. Sneyd
Mala Nataraj	The role of History in teaching Algebra	A/Prof. M. Thomas, A/Prof. A. Begg
Garry Nathan	Conceptual understanding in mathematics; Mathematical reasoning and argumentation	A/Prof. B. Barton, A/Prof. M. Thomas
Greg Oates	Technology and the Mathematics Curriculum	A/Prof. M. Thomas, A/Prof. B. Barton
Judith Paterson	Participation in professional development & its influence on pedagogical change for mathematics teachers	A/Prof B. Barton
Debasish Roy	Markov chain Monte Carlo algorithms, Bayesian Inference, Inverse problems	Dr. C. Fox
Sepideh Stewart	Conceptual understanding of linear algebra	A/Prof. M. Thomas, Prof. I. Reilly
Angela Tsai	Applied Mathematics – Numerical Analysis	Prof. J. Butcher, Dr. R. Chan

Inga Wang	Modelling the contraction of smooth muscle in the lung	Prof. J. Sneyd
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Masters Students

Name	Thesis Topic/Title	Supervisor(s)
Ku Azlina Ku Akil	Stability Analysis in solid body rotation flow. A linear operator theory approach	Dr. S. Wang
Matthew Auger	Torsion – free subgroups of small index in Coxeter groups associated with compact hyperbolic 4-manifolds	Prof. M. Conder
Tara Bonda	Research Portfolio	Prof. E. O’Brien
Robin Christian	Induced Non-Separating Cycles, Bridges and Connectivity	A/Prof. P. Bonnington
Susanne Kerr	How Year 13 students of Mathematics with calculus develop their understanding through interactions in the classroom	B. Kensington Miller
Miguel Jeffe	Functions on trees	Dr. D. Bryant
Viliami Latu	The development of the Tongan mathematical discourse	Dr. H. Bartholomew
Jing Ma	Modelling the distribution of volcanic ash	Dr. S. Taylor
John Christopher McCabe-Dansted	Approximability and computational feasibility of Dodgson’s rule	Dr. A. Slinko
Johanna McHardy	Investigating the impact of Technology Integrated Flexible Programme on Senior Mathematics Students’ Learning	A/Prof. M. Thomas
Lynda Vivien Merriman	Using media reports to develop statistical literacy in Year 10 students	Dr. M. Pfannkuch
Graham Reihana	Research Portfolio	A/Prof. J. Sirán
Vithya Shanmuganathan	The Phase error of explicit Runge-Kutta-Nystrom methods	Dr. P. Sharp
Maree Shaw	Mathematical techniques applied to Neuroscience	C. King
Elisabeth Sneddon	Collaborative Tutorials and the development of team skills in Engineering Mathematics	Dr. H. Bartholomew
Robert Stafford	Tractor Calculus and invariants for conformal sub-manifolds	A/Prof. R. Gover
Joanne Woodward	Statistical thinking as modelled by two Year 10 teachers when teaching informal inference using box plots	Dr. M. Pfannkuch
Cathy Wu	Comparative Taiwanese and NZ students’ algebraic understanding	A/Prof. M. Thomas

Honours Students

Name	Thesis Topic/Title	Supervisor(s)
Michael Brough	Seifert surfaces and knot invariants	Prof. D. Gauld
Wen Duan	Global stability analysis for the solid body rotation flow: complete analysis of the seventh and eighth order approximation	Dr. S. Wang
Emily Harvey	An investigation into canard solutions in multiple timescale systems	Dr. V. Kirk
Shaun Hotchkiss		
Dawoomi Kim	Stochastic partial differential equations: the effect of noise	Dr. P. Sharp
Simon Marshall		A/Prof. J. Siran
Stephanie Ng	Understanding PISA 2003: student achievement between and within schools and what an international perspective can say about New Zealand education	Dr. M. Pfannkuch
Oscar Parkyn	The phase error of explicit Runge-Kutta Nyström methods	Dr. P. Sharp
Sameeha Qaiser	A bifurcation analysis of a mathematical model with multiple time-scales	Dr. V. Kirk
Sammy Shen	The long term survival of asteroids	Dr. P. Sharp
Callum Sleigh	Harmonic analysis on locally Abelian groups	Dr. T. ter Elst
David Yu	Close approaches of an asteroid	Dr. P. Sharp

(b)

Research Fellows

Dr. Hyuck Chung
Dr. Laura Ciobanu
Dr. Richard Evans
Dr. Ye Yoon Hong
Dr. Edward (Shih-Chang) Huang
Dr. Abdul Mohammad
Dr Paul-Andi Nagy
Dr. Bart Oldeman
Dr. Caroline Poisard

Research visitors to the Department of Mathematics

Research Visitors

Prof. Abdolaziz Abdollahi
Prof. Murali Agastya
Dr. Keng-Cheng Ang
Prof. Boris Altshuler
Dr. Mathieu Baillif

Affiliation

Shiraz University, Iran
University of Sydney
Nanyang Technological University
Columbia University, NY
University of Geneva

Prof. Deborah Ball	University of Michigan
Prof. Hyman Bass	University of Michigan
Dr. Rick Beatson	University of Canterbury
Mr. Ronald Begg	University of Canterbury
Prof Len Bos	University of Calgary
Dr. Walter Bossert	University of Montreal
Prof. Peter Brooksbank	Bucknell University
Dr. Henning Bruehn	Hamburg University
Prof. Jochen Brüning	Humboldt University of Berlin
Prof. Jon Carlson	University of Georgia
Prof. Maria do Carmo Domite	University of Sao Paulo
Dr. Greg Chini	University of New Hampshire
Dr. Louise Parsons Chini	University of New Hampshire
Dr. Simon Chiossi	Humboldt University of Berlin
Prof. Sean Cleary	The City University of New York
Prof John Conway	Princeton University
Dr. Heiko Dietrich	Technical University Braunschweig
Prof. Michael Eastwood	University of Adelaide
Prof Bettina Eick	Technical University Braunschweig
Prof. Pavel Exner	Doppler Institute, Prague
Dr. Roger Fenn	Sussex University
Prof. Paul Fong	University of Illinois at Chicago
Dr. Steven Galbraith	Royal Holloway, University of London)
Dr. Georg Gottwald	University of Sydney
Prof. John Grue	University of Oslo, Norway
Dr. Colin Guillarmou	University of Nice and the ANU
Prof. Richard L. Hall	Concordia University, Montreal, Canada
Dr. Thierry Hild	Université de Fribourg (Switzerland)
Dr. Frank Himstedt	Munich University of Technology
Prof. Derek Holton	Otago University
Dr. Ville Kolehmainen	University of Kuopio, Finland
Prof. Ruth Kellerhals	University of Fribourg (Switzerland)
Dr. Steven Kennewell	Swansea University, UK
Prof Robert Jajcay	Indiana State University
Prof. Shouli Jiang	Shandong University
Prof Jari Kaipio	University of Kuopio, Finland
Dr. Ernie Kalnins	University of Waikato
Dr. Jonathan Kress	University of New South Wales
Prof. Bernd Krauskopf	University of Bristol
Dr. Bernhard Kroen	Hamburg University
Dr. Kate Lee	Queensland University of Technology
Dr. Dmitry Malinin	USP Fiji
Dr Ben Martin	University of Canterbury
Sione Ma'u	Indiana University
Prof. Alan McIntosh	The Australian National University
Prof. Mark McGuinness	Victoria University of Wellington
Prof. Abdul Mohamad	Sultan Qaboos University
Mark Nelson	University of Wollongong and Australian Defence Force Academy
Prof. William Newman	UCLA
Julia Novak	Royal Holloway, University of London
Prof. Peter Nyikos	University of South Carolina
Prof. John OCallaghan	Australian National University
Prof. Helcio R. B. Orlande	Federal University of Rio de Janeiro
Dr. Hinke Osinga	University of Bristol
Dr. Konstantin Pankrashkin	Humboldt-University of Berlin
Dr. Malte Peter	University of Bremen
Dr. Robyn Pierce	Ballarat University, Australia
Dr. Helmut Podhaisky	Martin-Luther-University Halle-Wittenberg
Prof. Derek W. Robinson	Australian National University
Dr. Colva Roney-Dougal	University of St Andrews
Prof. Bruce Richter	University of Waterloo
Prof. Derek W. Robinson	Australian National University

Prof. Neil Roberston
 Prof. Bjorn Sandstede
 Prof. Norbert Seifert
 Harvi Sidhu

Dr. Emily Stone
 Dr. Peter A. Streufert
 Dr. Izolda Sturova
 Dr. Gareth Vaughan
 Dr. Martin Wechselberger

Ohio State University
 University of Surrey
 University Leoben, Austria
 University of Wollongong and
 Australian Defence Force Academy
 University of Western Ontario
 University of Montana - Missoula
 Lavrentyev Institute of Hydrodynamics, Russia
 Otago University
 University of Sydney

(c) Research Grants.

MARSDEN GRANTS

Grantee	Amount	Title
Prof M. Conder, A/Prof. J. An, Prof. E. O'Brien	169525	Group actions, representations, structure and algorithms
Prof. D. Gauld, Prof. S. Deo (Allahabad) (AI)	346667	Dynamics on non-metrisable surfaces and manifolds

OTHER EXTERNAL GRANTS

Grantee	Amount	Title	Funding Body
A/Prof. B. Barton	5400	Indigenous mathematics teachers international research project	Royal Society of NZ (ISAT)
A/Prof. B. Barton, J. Paterson	50000	Teachers Learning Mathematics	Teaching and Learning Research Initiative
Dr. H. Bartholomew	5000		Tertiary Education Commission
Dr. S. Greenwood		Pangarau – Achieving in Mathematics	New Zealand Institute for Mathematics and their Applications
Dr. C. McMurchy	2500		
Pilkington			
A/Prof. P Bonnington	1766685	BeSTGRID: broadband- enabled science and technology GRID	Tertiary Education Commission
Dr R P K Chan	1800	Visit by Prof Wu from Nanjing University	NZIMA
Prof. M. Conder	1835000	NZIMA - Centre of Research Excellence in the Mathematical Sciences	Centres of Research Excellence Fund (Tertiary Education Commission)
Prof. R. Fenn, Prof. D. Gauld, Prof. V. Jones	398000	Geometric methods in the topology of 3-manifolds	NZIMA
Dr. V. Kirk	2889	Multiple time-scale analysis of mathematical models of calcium oscillations and waves	Royal Society of NZ (ISAT)
Dr. M. Meylan, Dr. Izolda Sturova	2960	Time Dependent linear waves	Royal Society of New Zealand

Dr. M. Meylan	2500	Website for Water waves	NZIMA
Prof. E. O'Brien	2000	Primitive matrix groups	Oberwolfach Forschungsinstitut
J. Paterson, G. Oates	75000	The Great Maths and Science Origami Show	Royal Society of New Zealand
Dr. A. Slinko	2667	Adaptive choice in complex environments	Royal Society of NZ (ISAT)
Dr. A. Slinko	5000	Visit by Prof. Mamoru Kaneko (Japan)	Economic Design Network (ARC)
A/Prof. M. Thomas	1000	Advanced mathematical thinking	Otago University
A/Prof. M. Thomas	5500	Neuropsychological aspects of mathematical thinking	Texas Instruments
A/Prof. M. Thomas	1100	Advanced mathematical thinking	Otago University

UNIVERSITY OF AUCKLAND STAFF RESEARCH GRANTS

	Amount	Title
Dr. H. Bartholomew Dr. S. Greenwood Dr. C. McMurchy Pilkington	17000	Pangarau – Achieving in Mathematics
A/Prof. R. Gover	55000	Geometric invariants for conformal and related structures – University of Auckland Postdoctoral Fellowship
A/Prof. R Gover	49100	Invariants, special structures, etc. – University of Auckland Research Fund
Prof. B. Pavlov	5400	Resonance Spin Filter
A/Prof. M. Thomas S. Stewart	3500	Communicating linear algebra: New Representations using the technology
Prof J. Sneyd	121233	Modeling smooth muscle contraction in the lung
Dr. M. Pfannkuch	13200	Developing students' statistical thinking
Dr. T. ter Elst	13632	Global analysis of elliptic operators
Dr. M. Meylan	10000	Measurement of wave attenuation in the marginal ice zone
Dr. P. Sharp	1750	Chandler's wobble and Earth's long-term climate
Dr. M. Meylan, A. Kohout	2800	Wave scattering in the marginal ice zone

VI. STAFF LEAVE AND CONFERENCES

Leave

Name	Type	When	Where
A/Prof. J. An	other		
Dr. H. Bartholomew	other	February 2006 – March 2006	King's College, London
A/Prof. B. Barton	other	6 May – 2 June August 2006-	UBC, Canada, Open University, UK, UC Berkeley, USA
Dr. L. Ciobanu	Long	August 2007 April to July	CRM Barcelona
Prof. M. Conder	short	2006	Slovenia, Canada, UK, Portugal, Switzerland
Prof. D. Gauld	Long	Semester 1	Melbourne, Allahabad, Auckland
A/Prof. R. Gover	Long	Semester 1	Austria, Czech republic, France, USA
H. McKenzie	short	July	Australia
H. McKenzie	short	13Dec-4 Jan	Australia
Dr. A McNaughton	other		
Prof. E. O'Brien	other	December July - August	University of Western Australia
Prof. E. O'Brien	other	2006	Oberwolfach and London
J. Paterson	Long	January to June	New Zealand, U.S.A., England, South Africa
Dr. P. Sharp	other	January 9 - 29	Canada
A/Prof. J. Sirán	short	January	Slovakia
A/Prof. J. Sirán	short	April	Peking, China
Dr. A. Slinko	short	Semester 2 22 November -	Montreal (Canada)
Dr. T. ter Elst	other	20 December 27 June - 15	Ulm, Germany
Dr. T. ter Elst	short	July	ANU, Canberra

Conferences

Conference Contribution - Oral Presentation

BARTON, B., REILLY I., 'Topology and Language', *Third International Conference on Ethnomathematics*, Auckland, 12-16 February 2006

BARTON, B., SNEDDON, J. & NEVILLE-BARTON, P. If "ifs" and "ans" were pots and pans ... we'd have a lot more thinkers: English Second Language Speakers Learning Undergraduate Mathematics. *NZ Mathematics Colloquium*, Waikato University, December, 2006.

BRÜNING J., PAVLOV B., 'Remark on calculation of the Kirchhoff coefficients for Helmholtz resonator', *Operator Theory and Mathematical Physics (OTAMP 2006)*, Lund, 15-22 June 2006, 13 p

CALVERT, B. D., 'Uniqueness of solutions to a superlinear three-point boundary value problem', *AIMS*, Poitiers, France, 25-28 June 2006

CAO, J. 'Baire versus Volterra spaces', *International Conference in analytic topology and its applications*, Rotorua, New Zealand, June 2006

CAO, J.. 'Baire spaces, Tychonoff powers and the Vietoris Topology', *10th Prague Topology Symposium*

GOVER, R., 'Invariant Prolongation and Detour Complexes' at the workshop 'Symmetries and Overdetermined Systems of Partial Differential Equations' 2006 IMA Summer Program, Institute for Mathematics & Its Applications, Minnesota USA

GOVER, A.R. 'CR tractors and the Fefferman metric', *AMSI workshop on Several Complex Variables and CR geometry*, University of New England, Armidale, NSW, Australia, 6 -- 10 February, 2006

GOVER, A.R. 'Invariant Prolongation and Detour Complexes', *Symmetries and Overdetermined Systems of Partial Differential Equations*, Institute for Mathematics & Its Applications, Minnesota, July 17-August 4, 2006

GOVER, A.R. 'Prolonged differential systems and a holonomy characterisation of the Fefferman Space', *The 8th Pacific Rim Geometry Conference*, Australian National University, December 11 to December 15, 2006

HEARD, A.D. 'Stability of adaptive numerical algorithms', *NZ Mathematics Colloquium 2006, University of Waikato, Hamilton, 4-6 December, 2006*

O'BRIEN, E. 'Using involution centralisers in linear group algorithms', Invited lecture, Asymptotic Group Theory", Jerusalem, May 2006

O'BRIEN, E. 'Recent progress on matrix group recognition', Invited lecture, "Computational Group Theory meeting", Oberwolfach Forschungsinstitut, July 2006

O'BRIEN, E. 'Taming the tiger: reflections on algorithms for linear groups', Invited lecture, "Groups and Computation", London, July 2006

MOORS W. 'Metrizability of compact convex sets', *International Conference on Analytic Topology and its Applications*, Rotorua, New Zealand, 10-14 July 2006

MOORS W. 'Metrizability of compact convex sets', Plenary talk, 10th Prague Topology Symposium, Prague, Czech Republic, August 2006

PAVLOV B. 'Solvable model for Helmholtz Resonator, Dynamic Systems', *Differential Equations and Applications. Sixth International conference AIMS*. Poitiers, 25-28 June 2006.

PAVLOV B. 'Dynamic Systems, Differential Equations and Applications', *Nanosea: International conference of surface science*, Aix-en-Provence, 30 June- 5 July 2006.

POISARD, C. 'Studying mathematical objects in class, the example of the Chinese abacus', *Third International Conference of Ethnomathematics, Auckland, 12-16 February 2006, 2006*

SIRAN J., 'Hyperbolic geometry from an algebraic point of view', Workshop Geometry and its Applications, Slovak Univ. Technology, 2-02-2006

SIRAN J., 'Regular maps on a given surface', 31st Australasian Conf. in Combinatorial Math and Combinatorial Computing, Australia, 8-07-2006

SIRAN J., Regular maps on a given surface - a survey, *6th Czech-Slovak International Symposium on Graph Theory, Combinatorics, Algorithms and Applications*, Czech Rep., 15-7-2006;

STEWART, S. 'Process-object difficulties in linear algebra: Eigenvalues and eigenvectors', *The 30th Conference of the International Group for the Psychology of Mathematics Education*, Prague, Czech Republic, July, 2006

STEWART, S. 'Student thinking about eigenvalues and eigenvectors: Formal, symbolic and embodied notions', *The 29th annual Conference of the Mathematics Education Research Group of Australasia: MERGA.*, Canberra, Australia, July, 2006

Conference Contribution - Conference Paper

BUTCHER, J.C. 'General linear methods for ordinary differential equations', *6th PanAmerican Workshop on Applied and Computational Mathematics, Oaxaca, Mexico, 23-28 July, 2006*

BUTCHER, J.C. 'Order and stability for general linear methods', *11th Seminar "NUMDIFF"*, Halle, Germany, 4-8 September, 2006

BUTCHER, J.C. 'Order and stability for general linear methods', *NZMC 2006, Hamilton, New Zealand, 4-6 December, 2006*

BUTCHER, J.C. 'Scientific Computation and differential equations', *Annual Fundamental Science Seminar, Malaysia*, 6-7 June, 2006

BUTCHER, J.C. 'Some examples of structure preservation', *ANZIAM Conference 2006, Mansfield, Victoria, Australia*, 5-9 February, 2006

CHUNG, H., EMMS, G., DODD, G., MCGUNNIGLE, K. 'Improving the Impact Insulation of Light Timber Floors', *WESPAC IX 2006, Seoul, Korea*, 26 -28 June, 2006

CHUNG, H., FOX, C. 'Low-frequency vibration of light-weight floor/ceiling structures with irregularly shaped joist beams', *Inter-Noise 2006, Honolulu, Hawaii, USA*, 3-6 December, 2006\

CONDER, M.D.E. 'A more detailed classification of symmetric cubic graphs', *International Workshop on Symmetries of Combinatorial Structures, Ljubljana (Slovenia)*, May, 2006

- CONDER, M.D.E. 'A more detailed classification of symmetric cubic graphs', *SIAM Conference on Discrete Mathematics, Victoria, BC (Canada)*, June, 2006
- CONDER, M.D.E. 'A more detailed classification of symmetric cubic graphs', *Workshop on Infinite Aspects of Topological Graph Theory, Auckland*, February, 2006
- CONDER, M.D.E. 'Chiral polytopes', *3rd Conference on Symmetries of Graphs, Maps & Complexes, Aveiro (Portugal)*, July, 2006
- CUI, T., FOX, C., NICHOLLS, G., O'SULLIVAN, M.J. 'Bayesian Inference for Geothermal Model Calibration', *New Zealand Geothermal Workshop, Auckland*, 15th November, 2006
- FOX C., B. PAVLOV, PETROVA L., 'Saint-Venant principle and a solvable model for the thin plate under boundary strain', *New Zealand Math. Colloquium*, 3-5 Dec. 2006, Hamilton, 33p.
- GOVER, A.R. 'Invariant Prolongation and Detour Complexes', *Symmetries and Overdetermined Systems of Partial Differential Equations, Institute for Mathematics and its Applications, Minnestoa*, July 17-August 4, 2006
- HARMER M., PAVLOV B., YAFYASOV A., 'Boundary conditions at the junction', *International Workshop on Computational Electronics (IWCE-11), Vienna*, 25 May - 29 May 2006, .IWCE-11 , book of abstracts, pp 241-242. To appear in "Computational Electronics"
- MCCABE-DANSTED, J.C., PRITCHARD, G., SLINKO, A.M. 'Approximability of Dodgson's Rule', *VIII International Meeting of the Society for Social Choice and Welfare, Istanbul*, July 13-17, 2006, https://gemini.econ.umd.edu/cgi-bin/conference/conference.cgi?action=login&db_name=IIMSCC
- OATES, G., PATERSON, J.E., BAXTER, J., GRIBBEN, H. 'The Great Origami Maths and Science Show What's up in Origami and Mathematics Down-under?', *The Fourth International Conference on Origami in Science, Mathematics and Education (4OSME)*, Caltech, Pasadena, California, 8-10 September, 2006
- PETER, M.A., MEYLAN, M.H., LINTON, C.M. 'Scattering of Water Waves by Arrays of Arbitrary Bodies.', *International Workshop on Water Waves and Floating Bodies, Loughbourogh, UK*, 2-5 April 2006, 2006
- PAVLOV B., YAFYASOV B., 'Standing waves and resonance transport mechanism in quantum networks', *Differential Equations and Applications. Sixth International conference AIMS. Poitiers*, 25-28 June 2006, Accepted for publication in "Surface Science". Dynamic Systems, 9p.
- SIRAN, J. 'Invited address: Regular maps on a given surface: A survey', *6th Czech and Slovak International Symposium in Graph Theory, Prague*, 10-15. 6. 2006, 2006
- SIRAN, J. 'Invited talk: Hyperbolic geometry from the algebraic viewpoint', *Workshop in Geometry, Bratislava*, 2. 2. 2006, 2006
- SIRAN, J. 'Invited talk: Regular maps on surfaces', *Australian Workshop on Combinatorics and algorithms, Alice Springs*, 7.-12.7.2006, 2006
- SIRAN, J. 'Pinched regular maps', *Topological aspects of infinite graphs, Auckland*, February 13-17, 2006
- SLINKO, A.M., WHITE, S.M. 'On the Manipulability of Proportional Representation', *VIII International Meeting of the Society for Social Choice and Welfare, Istanbul*, 13-17 July, 2006, https://gemini.econ.umd.edu/cgi-bin/conference/conference.cgi?action=login&db_name=IIMSCC
- TEE G. J. 'A tricky Tripos problem- generalised'. 2006 NZ Mathematics Colloquim, Waikato University, 3 to 5-12-2006, 2006.
- TER ELST T. 'Regularity of elliptic operators', *Heat kernels in mathematics and physics, Conference, Blaubeuren, Germany*, 28-11-2006 - 2-12-2006

Seminars given by Staff, within other Universities or Teachers' Associations

- BARTHOLOMEW. H. 'Emotion and Mathematics Professional Development in a Low Socio-Economic Area in New Zealand', *King's College, London*, 31-01-2006
- CONDER M., Colloquium lecture, Institute of Mathematics & Physics, University of Ljubljana (Slovenia), May 2006
- CONDER M., Seminar at University of Fribourg (Switzerland), July 2006.
- GOVER R, 'Yang-Mills detour complexes and conformal geometry', invited lecture University of Washington, USA. 9-03-2006

GOVER R , 'Detour complexes and a generalisation of Q-curvature', invited lecture Universite Henri Poincare, Nancy, 1-06-2006

GOVER R , 'Einstein metrics, conformal geometry, and tractor calculus', invited lecture Seoul National University, Korea, 15-11-2006,

GOVER R , 'Prolonged differential systems and Detour Complexes', invited lecture Seoul National University, Korea , 17-11-2006

GOVER R , 'CR tractors and the conformal Fefferman space', Seoul National University', invited lecture, Korea, 20-11-2006

GOVER R , 'Prolonged differential systems and a holonomy characterisation of the Fefferman Space', invited lecture, University of Warsaw, 23-11-2006

GOVER R , 'Einstein metrics, conformal geometry, and tractor calculus', invited lecture , University of Warsaw, 5-12-2006

MOORS, W. 'The product of a Baire space with a hereditary Baire metric is Baire', University of Western Australia, Miniconference on Functional Analysis

MOORS, W. 'Contributed Baire spaces and the Vietoris topology', University of Western Australia, Miniconference on Functional Analysis

OATES, G. 'Mathematics Tutoring at Auckland University', Department of Mathematical Sciences Mathematics Education Seminar, University of Wisconsin-Milwaukee Mathematical Sciences Department, 30-08-2006

OATES, G. 'Origami Revisited: Connecting your students to learning mathematics.', Special workshop for teachers of mathematics and mathematics faculty., University of Wisconsin-Milwaukee Mathematical Sciences Department, 31-08-2006

OATES, G. 'The Great Origami Maths and Science Show: What's up in Origami and Mathematics Down-under?', Department of Mathematical Sciences Colloquium, University of Wisconsin-Milwaukee Mathematical Sciences Department, 30-08-2006

OATES, G. 'Folding the Curriculum', *Bay of Plenty Mathematics Association Mathematics Conference, Rotorua Convention Centre, 24 November, 2006*

OATES, G. 'Web-based free software resources for Trigonometry', *Mathematics with Calculus, The University of Auckland, 1 December, 2006*

OATES, G., 'Using Origami to meet curriculum objectives', *Bay of Plenty Mathematics Association (BOPMA) Mathematics Conference*

O'BRIEN E. A series of 14 lectures on 'Computational Group Theory', ICE-EM/AMSI Summer School: Melbourne, February 2006

O'BRIEN E. 'Involutions in linear groups', University of Western Australia, December 2006

PARNELL, S. M. 'Practical Ideas and Activities in Mathematics', *Bay of Plenty Mathematics Association Mathematics Conference, Rotorua, New Zealand, 24 November, 2006*

PATERSON, J.E. 'Doing and Undoing - developing students' confidence in checking their own work in Mathematics', *Bay Of Plenty Mathematics Association Teachers' Day, Rotorua, 24 November, 2006*

PATERSON, J.E. 'Misconceptions in Mathematics - students' and teachers', *University of Auckland Calculus Day, University of Auckland, Tamaki Campus, 1st December, 2006*

PATERSON, J.E. 'A workshop for teachers: Looking at learning about encryption.', *Milwaukee Mathematics Project (MMP)* funded by the National Science Foundation, University of Milwaukee, Wisconsin, 20-05-2006

PATERSON, J.E. 'Probability and Statistics for the South African Classroom', *AIMSSEC, Stellenbosch, South Africa, 23-06-2006*

PATERSON, J.E. 'Using Mathematics to encourage teachers' discussion of learning and teaching', University of Milwaukee, Wisconsin, 19-05-2006

POISARD, C. 'Bones to multiply and multiplication algorithms'. *BOPMA Conference. Rotorua, New Zealand, 24 November 2006.*

SIRAN J., Classification of regular maps of Euler characteristic, Peking University, April 2006.

SLINKO A., 'Parametrised complexity and aggregation of rankings', Invited fully funded lecture Discrete Mathematics Seminar of Carleton University (Ottawa);

SLINKO A., Fully funded colloquium lecture "Combinatorics of Comparative Probability Orders" - Department of Mathematics University of Ottawa;

THOMAS M. J., 'Developing Versatility in Mathematical Thinking at Retirement as Process and Concept: A Festschrift for Eddie Gray and David Tall', Invited plenary talk, Charles University, Prague, Czech Republic;

THOMAS M.J., 'Mathematical Thinking and the Graphic Calculator', Invited talk, Mathematics Department, Otago University;

THOMAS, M. O. J. 'Conceptual Representations and Versatile Mathematical Thinking'. *Proceedings of ICME-10, the International Congress of Mathematical Instruction, Copenhagen, Denmark*, . (In print)

TER ELST T. 'Degenerierte Operatoren', Invited lecture, University of Ulm, 11-12-2006

VII. COMMUNITY SERVICES

Public Lectures: Two public lectures in the series of Mathematics Public Lectures were held during the year. The first public lecture given by Prof John Conway (Princeton University) was titled "Tangles, bangles and knots". In October, the second lecture was given by Dr Philip Sharp, a senior lecturer in the Department of Mathematics. The title of the lecture was "Near Earth Objects: threats and opportunities". Both were well attended by both members of the university and the general public.

The Great Mathematics and Origami Show. Supported by a grant from the Royal Society of New Zealand Several members of the department were involved with "The Great Origami Mathematics and Science Show" culminating in a tour of schools throughout New Zealand. Hugh Gribben gave an interview on the show for National Radio.

A/Prof J. An: Associate Editor of New Zealand Journal of Mathematics; Treasurer of the New Zealand Journal of Mathematics.

A/Prof. B. Barton: Assistant Editor Educational Studies in Mathematics; Vice-President, International Commission on Mathematical Instruction; Editorial Board: For the Learning of Mathematics; Pythagoras. Convenor of the Third International Conference of Ethnomathematics, Auckland, 2006.

A/Prof P. Bonnington: Chief Managing editor – Australasian Journal of Combinatorics.

Prof. J. Butcher: Editor Board Member for: Applied Numerical Mathematics, Numerical Algorithms, Journal of Universal Computer Science, Acta Numerica

A/Prof. B. Calvert: Editor Board Member for the journals CSSP and AAA.

Dr. R. Chan: Organiser of Calculus Day for secondary teachers.

Dr. L. Ciobanu: Organiser, Mathematics Colloquium, University of Canterbury, March 2006.

Prof. M. Conder: Member of RSNZ Council; National Science Panel; Grafton Hall Board; Advisory Group for the Strategic Review of Research in the Mathematics Sciences in Australia; Co-director (and Treasurer) of NZ Mathematical Research Institute (Inc.); Member (and President-Elect) of the Council of the Academy of the Royal Society of NZ; Editorial Board of NZ Journal of Mathematics.

Prof. D. Gauld: New Zealand Mathematics Research Institute (founding secretary); New Zealand Journal of Mathematics Committee (chair); Member of UASC and of the Education Faculty Staffing Committee

A/Prof. R. Gover: Editorial Board Member for the journals "New Zealand Journal of Mathematics" and "Symmetry, Integrability and Geometry: Methods and Applications"; University of Auckland Rhodes Scholarship selection panel; Coaching and mentoring of Rhodes applicants.

Dr. A. Heard: Secretary/Treasurer of NZ branch of ANZIAM; Role model and mentor for women students.

Dr. V. Kirk: Organiser, 2-day workshop for Women in Mathematical Sciences, to be held in November 2007. Funded by NZIMA; Member, NZMS; ANZIAM; SIAM, including activity group on Dynamical Systems; Programme Committee, NZIMA thematic programme on Dynamical Systems and Numerical Analysis.

Dr. B. Miller-Reilly: New Zealand contact for the Australasian Bridging Mathematics Network; The New Zealand agent for Adults Learning Mathematics.

Dr. W. Moors: Organiser of the international conference "International Conference on Analytic Topology and its Applications" Rotorua, 10-14 July 2006.

Dr. P. A. Nagy: Co-organiser of the international workshop “Special Geometries in Mathematical Physics”, Kühlungsborn, Germany, 12-18 March 2006.

G. Nathan: Contract work for Ministry of Education

G. Oates: Advisory Committee for 2007 Delta conference in Argentina.

Prof. E. O'Brien: Member of Editorial Board, J of Algebra (Computational Algebra); Editor (joint with Derek Holt, Warwick) for special issue of J. Algebra, which appeared June 2006; Member of Editorial Board, J of the Australian Mathematical Society (Group Theory and Computational Algebra); Grant assessor for a number of international funding agencies.

S. Parnell: Contract work for NZQA.

Dr. M. Pfannkuch: Member of New Zealand Statistical Association education committee; Member of New Zealand Mathematics Curriculum Reference Group; Writer for 2008 Statistics school curriculum; New Zealand correspondent for the International Association of Statistical Education; Member of Joint IASE roundtable/ICMI study International Program Committee; International Research Forum on Statistical Reasoning, Thinking, and Literacy Advisory Committee; ICOTS-7 International Programme Committee; Elected Member of International Statistical Institute; Associate Editor – Statistics Education Research Journal; International Program Committee for the 2008 International Commission on Mathematics Instruction study; Topic convenor for the International Conference on Teaching Statistics 2006; Guest editor for special issue of the Statistics Education Research Journal.

Prof. I. Reilly: Esquire Bedell; Member of the International Steering Committee for the DELTA conferences; Chair of The NZ Mathematical Olympiad Committee; Liaison Officer of the NZ Study Centre of the University of California Education Abroad Program for 7 New Zealand Universities.

Dr. P. Sharp: Gave evening courses on Astronomy through the Centre for Continuing Education; Gave two talks at the FoS Incredible Science Day.

A/Prof. J. Sirán: Member of Editorial Board of J. Combinat Theory Ser. B, Australasian J. Combinat., Mathematica Slovaca, Acta Mathematica Univ. Comenianae.

Dr. A. Slinko: Member of New Zealand Mathematics Olympiad Committee (NZMOC); New Zealand Asia-Pacific Math Olympiad (APMO) Representative; Chairman of the Problem Selection Committee of Auckland Mathematical Olympiad. Founding member of the Society for Economic Design; Co-organiser of the seminar "Discrete Mathematics and Social Sciences"; Member of the Scientific Committee of The 3rd Pan-Pacific Game Theory Conference (October 20-22, 2006, Beijing); Member of the editorial board of the Journal of Generalized Lie Theory and Applications.

M. Statham: Moderation of Mathematics Course for MIT: Maths 144.201; Contract work for NZQA.

A/Prof. M. Thomas: Contract work for NZQA; Editor, New Zealand Mathematics Magazine; Member of the executive of Auckland Mathematics Association; Editorial Board of Mathematics Education Research Journal

Dr. S. Waldron: member of the Editorial board of the International Journal of Theoretical and Numerical Approximation (IJOTANA).

VIII. OTHER MATTERS

Staffing

Dr Philip Sharp was promoted over the Senior Lecturer bar.

Dr Sina Greenwood was promoted to Senior Lecturer.

Dr Allison Heard was promoted over the Senior Tutor bar.

Awards

Prof. Ivan Reilly: Officer of the New Zealand Order of Merit for services to Mathematics

Prof. James Sneyd: James Cook Research Fellowship awarded for research entitled “Modelling airway smooth muscle: from Mathematics to asthma”

A/Prof. Paul Bonnington: Faculty of Science Dean’s Award for Excellence in Teaching

Department Administration

Many staff members have made substantial contributions to the effective administration of the Department. Thanks are especially due to all the general staff and to the following staff for taking on key responsibilities in the Department:

Deputy Head of Department	Prof. James Sneyd, Prof. Eamonn O'Brien, A/Prof. R. Gover
Departmental Committee	A/Prof. Bill Barton, A/Prof. Rod Gover, Dr. Paul Hafner Dr. Vivien Kirk, Prof. Eamonn O'Brien, Daniela Rovere, Prof. James Sneyd, Dr. Philip Sharp
Department Staffing Advisory Committee	A/Prof. Rod Gover, Dr. Vivien Kirk, Prof. Eamonn O'Brien, Dr. Hannah Bartholomew, A/Prof. Josef Siran
Academic Committee:	Algebra & Combinatorics: A/Prof. Josef Siran Analysis & Geometry: A/Prof. Rod Gover Applied Mathematics: Dr. Vivien Kirk Mathematics Education: A/Prof. Mike Thomas Post-Graduate Coordinator: Dr. Philip Sharp Undergraduate Coordinator: Dr. Jamie Sneddon Teaching Coordinator: Dr. Paul Hafner PhD Coordinator: Dr. Mike Meylan
Research and Postgraduate Committee	Prof. Marston Conder, A/Prof. Rod Gover, Dr. Sina Greenwood, Dr. Philip Sharp, Prof. James Sneyd, A/Prof. Mike Thomas, Dr. Shayne Waldron
Computer Committee	A/Prof. Paul Bonnington, Prof. Eamonn O'Brien, Dr. Philip Sharp, A/Prof. Mike Thomas
Head of Applied Mathematics Unit	Dr. Vivien Kirk
Head of Mathematics Education Unit	A/Prof. Mike Thomas
Head of Tamaki Mathematics Group	Dr. Robert Chan
Directors of Graduate Studies PhD	Dr. Mike Meylan
Director of Graduate Studies Masters and Postgraduate Diploma	Dr. Philip Sharp
Teaching Coordinator	Dr. Paul Hafner,
Research Coordinator	Prof. Marston Conder
BTech (Industrial Mathematics) Coordinator	Dr. Steve Taylor
Enrolment, Examinations, Timetable Coordinator	Dr. Jamie Sneddon
Publicity Officer	Daniela Rovere, Chris King
Regulations/Handbooks	Dr. Jamie Sneddon, Daniela Rovere, Adina Nagy
EEO/EEo Representative	Dr. Sina Greenwood, Daniela Rovere
NZ Mathematical Society Correspondent	Dr. Garry Tee
Webmaster	Daniela Rovere, Adina Nagy

Overseas Students & Ad Eundum Admissions	Dr. Philip Sharp
Library Liaison Officer	A/Prof. Bruce Calvert
Convenor of Staff/Student Liaison Committee	Greg Oates
Summer Research Scholarships Coordinator	A/Prof. Josef Siran
Ethics Advisor	A/Prof. Mike Thomas
Organiser of the Undergraduate Laboratories	Dr. Allison Heard
Markers Coordinator	Roy Swenson
Department Research Report Series	Olita Moala
Safety Officer	Daniela Rovere
Seminars:	
Algebra, Geometry & Combinatorics	Prof. Eamonn O'Brien
Analysis	Dr. Shayne Waldron
Applied, Computational & Industrial	
Mathematics	Dr. Philip Sharp
Geometric Analysis	A/Prof. Rod Gover
Topology	Prof. David Gauld
MESS (Mathematics Education)	Dr. Maxine Pfannkuch
GRID (Mathematics Education)	
Faculty Representatives:	
Arts	A/Prof. J. Siran
Business and Commerce	Dr. Arkadii Slinko
Engineering	A/Prof. Bruce Calvert
Faculty of Science Academic Committee	A/Prof. Mike Thomas

University Committees

A/Prof. Bill Barton

Faculty of Science Staffing Committee

A/Prof. C. Paul Bonnington

University Review of Research Administration
University Research Administration Committee
University IT Strategy and Policy Committee
University IT Faculties Forum

Prof. Marston Conder

Chair of Advisory Board for Bioengineering Institute
Chair of University's Budget Committee
Vice-Chancellor's Senior Management Team
Capital Planning & Budgeting Committee
Governing Board of Maurice Wilkins Centre for Molecular Biodiscovery

Prof. David Gauld

Faculty of Education Staffing Committee
University Academic Staffing Committee

A/Prof. Rod Gover

University Academic Staffing Committee
Faculty of Science Marsden Panel

Dr. Vivien Kirk

Faculty of Science Staffing Committee.

Prof. Eamonn O'Brien

Faculty of Science IT Committee.

Dr. Philip Sharp

Faculty of Science PG Committee
CCE Advisory Group.

Prof. James Sneyd

Faculty of Science Research Committee

A/Prof. Michael Thomas

Faculty of Science Academic Committee
Faculty of Science Staff Student Consultative Committee
Faculty of Arts Research Committee.
University Academic Programmes Committee

New Zealand Journal of Mathematics

The New Zealand Journal of Mathematics is jointly produced by the Department and the New Zealand Mathematical Society. Apart from the prestige of producing such a journal a major advantage for the University is the number of international journals which the University receives in exchange for the New Zealand Journal of Mathematics.

The Editorial staff consists of Prof. Gaven Martin (Editor), Dr Jianbei An (Associate Editor) and Ms Betty Fong (Production Assistant). Members of the Department who have served on the Editorial Board are Professors John Butcher, Marston Conder, David Gauld, Vaughan Jones and the Department's representative on the Management Committee is Prof. Ivan Reilly.

Maintaining Teaching Quality

This is undertaken by a newly formed Departmental Academic Committee, which meets monthly.

IX. OVERALL COMMENTS ON WORK AND PROGRESS WITHIN THE DEPARTMENT

The 2006 Review of the Department of Mathematics begins with the following Introduction:

The Department of Mathematics can rightly claim to be New Zealand's strongest mathematics department, and is the only one to offer a comprehensive programme in pure mathematics—as such it is a national asset. The Department has many members with high international reputations. ... The annual output of research is impressive.

However, The Department faces two pressing challenges, namely declining student numbers and the imminent or actual resignation and retirement of several of its senior staff. The Department will have to respond effectively to both if it is to prosper in the future.

The changing staff profile has been commented upon in both the 2004 and 2005 Reports. We continue to lose staff due to planned retirements, but have been unable to replace them due to the decline in student numbers. The upward trend in enrolments in the first month of 2007 is a hopeful sign that we may be able to stop this erosion of expertise - an erosion that has reached a stage where adequate coverage of our subject is in danger.

The turnover in General Staff continued in the early part of 2006, but we are now back to our full complement so that the extra load for these staff members is over. All the academic staff highly value the work done by this team.

In 2006 our newly revised courses were trialed for the first time, with general success. We look forward, in 2007, to consolidating these courses, extending them through to the 300-level, and systematic introduction of improved, technology-enhanced, delivery mechanisms. I hope, in my next annual report, to be able to comment on the new tutorial facilities that we have sought for so long.